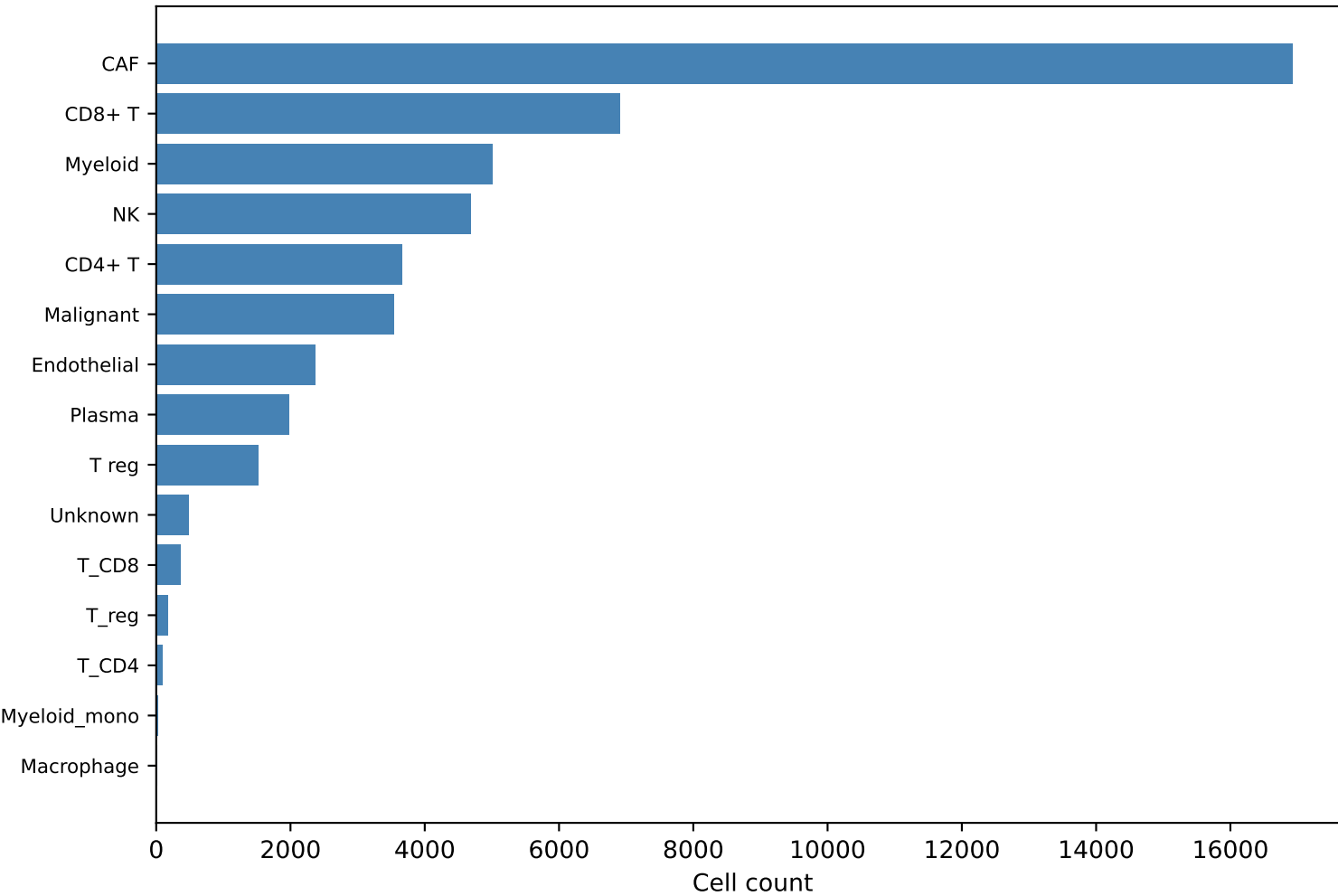
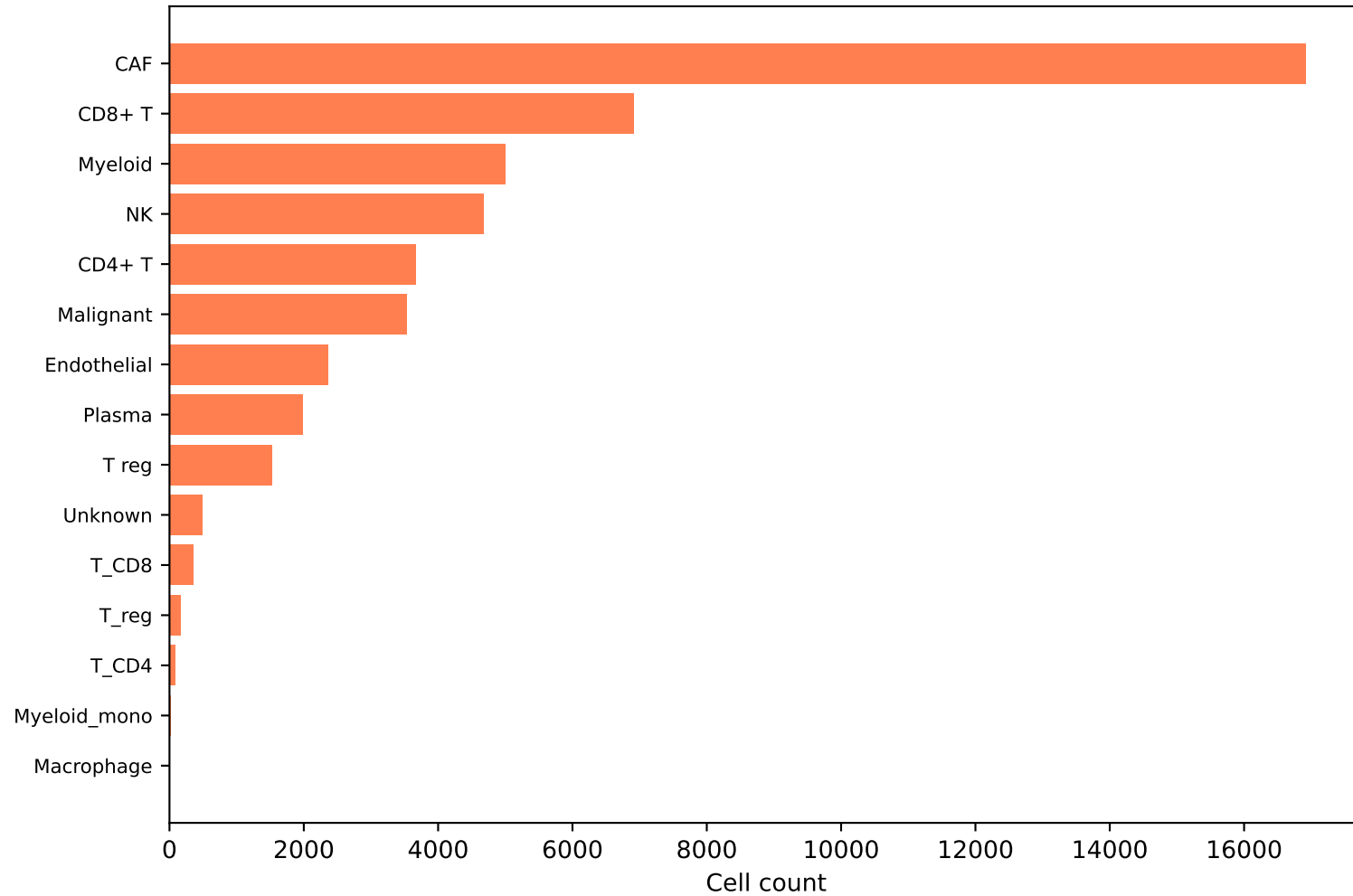


Finalized Cell Type Annotations

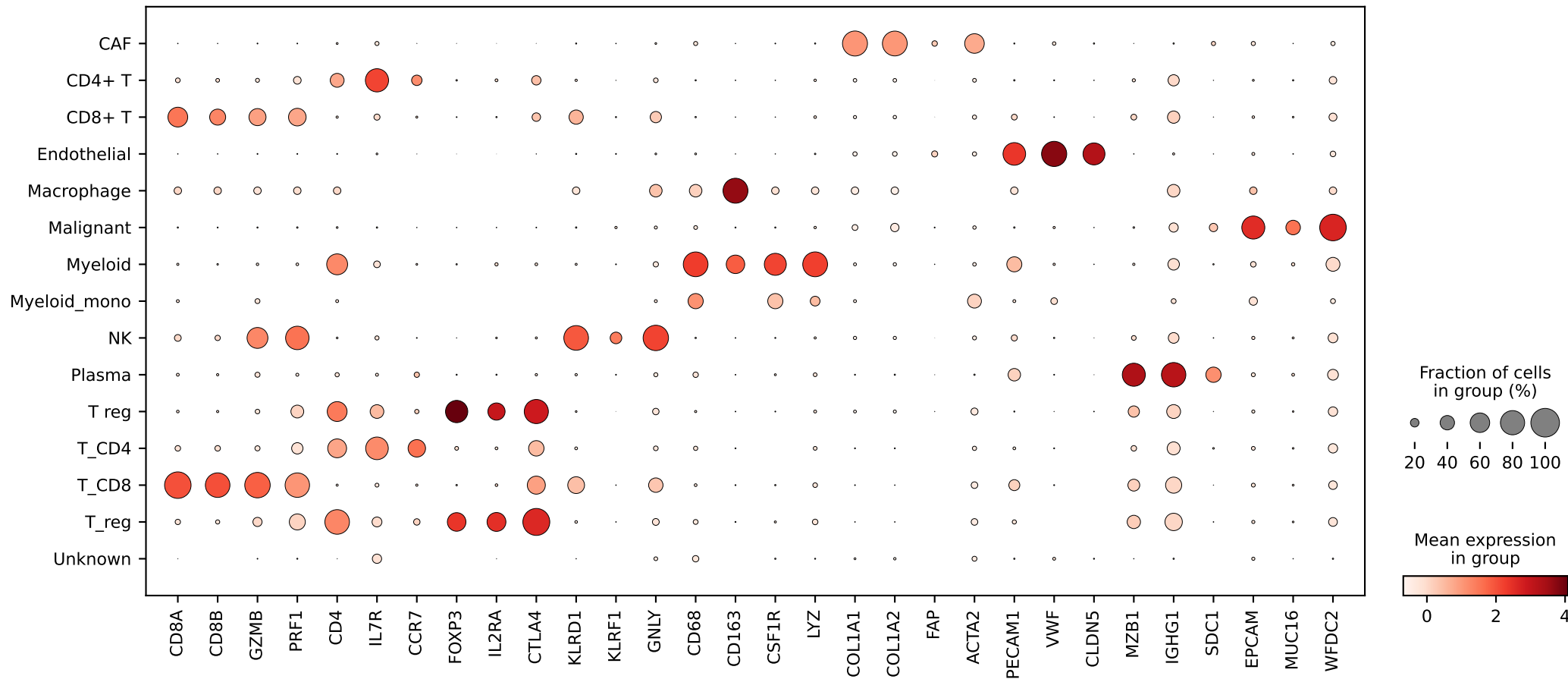
Cell type distribution



Cell type major distribution

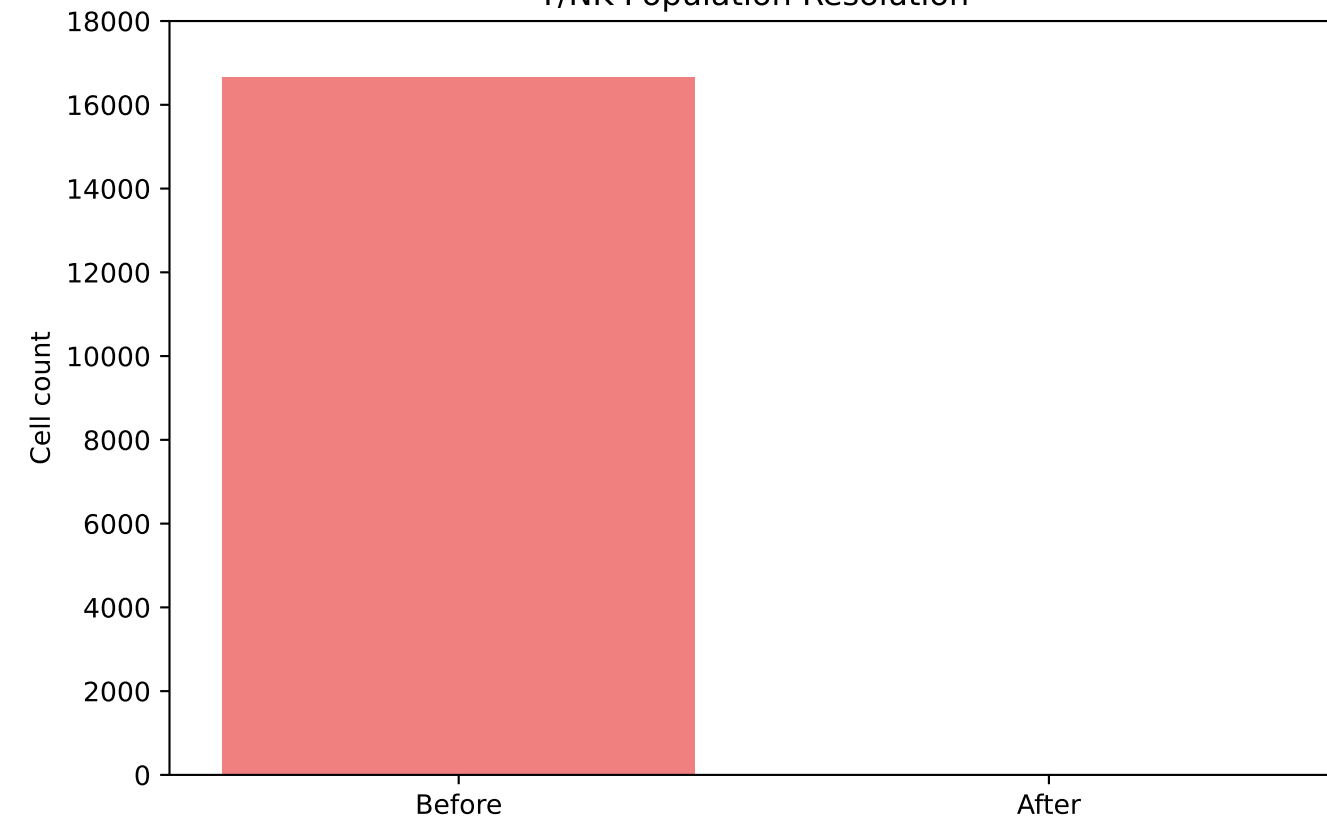


Marker Expression by Cell Type

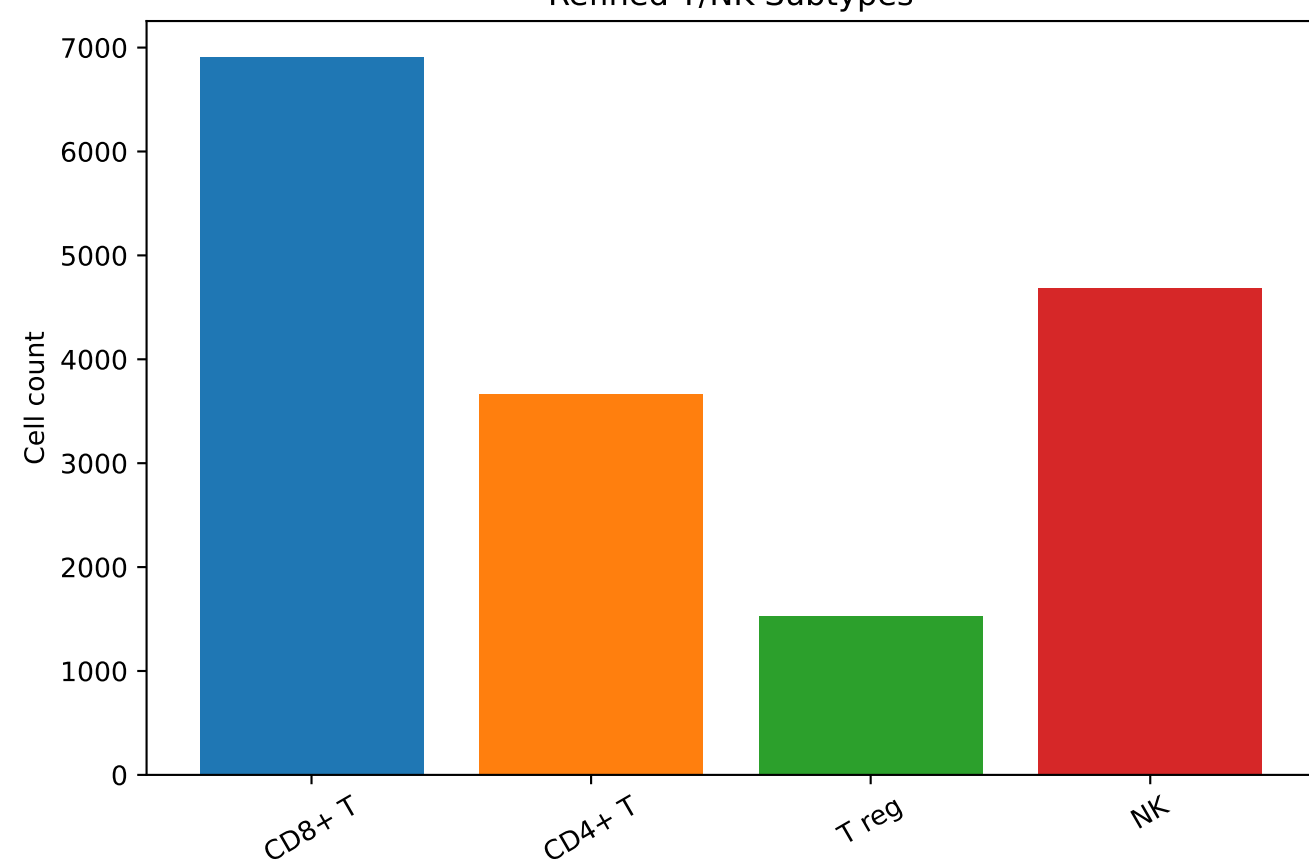


T/NK Sub-clustering Results

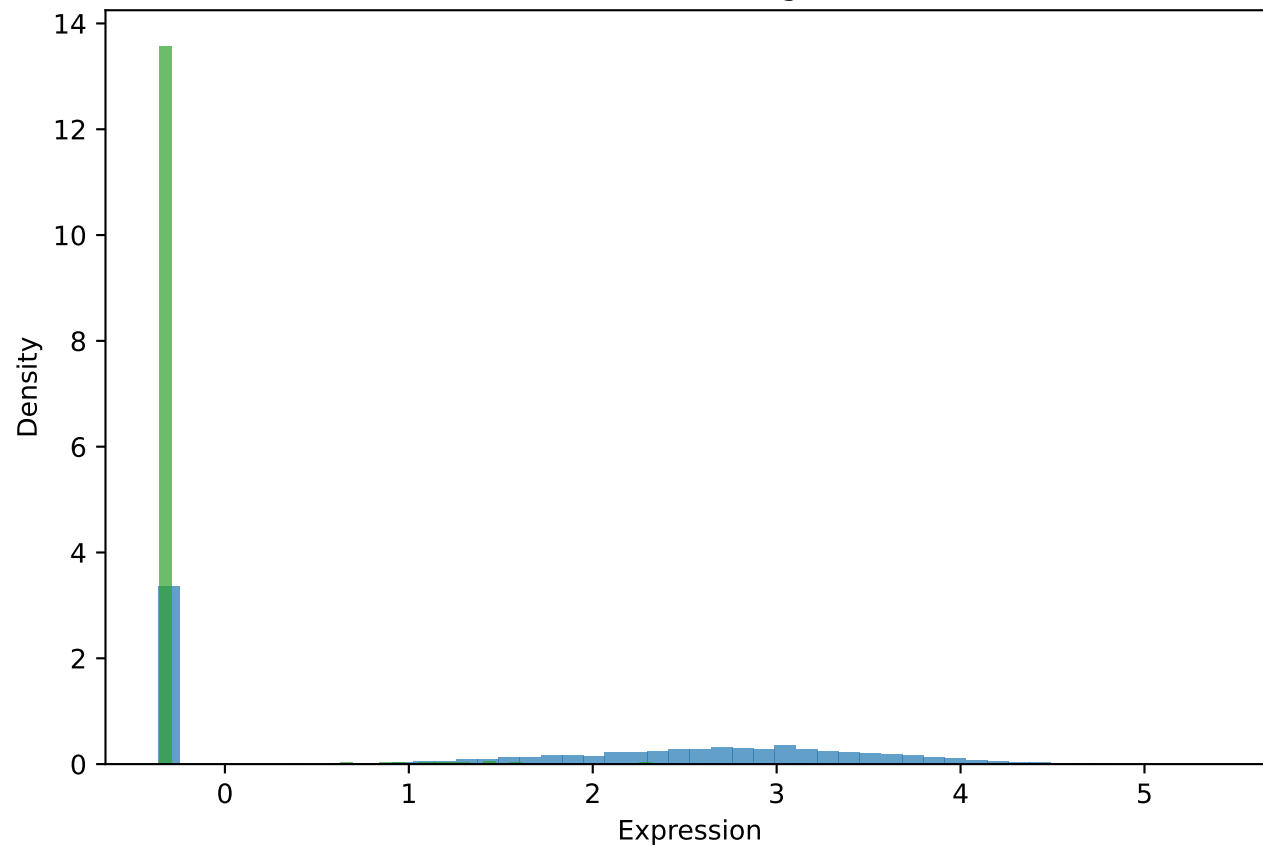
T/NK Population Resolution



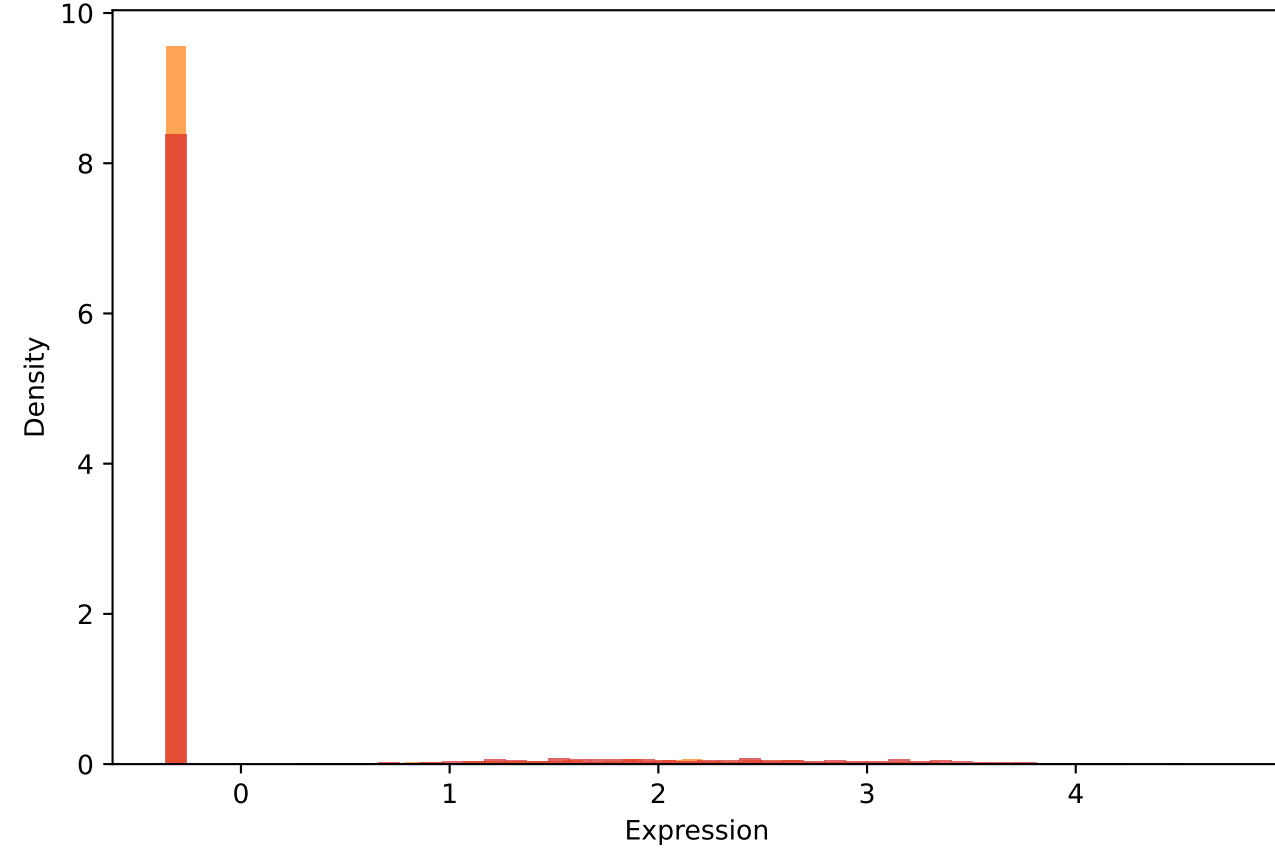
Refined T/NK Subtypes



CD8A in T reg



CD8A in NK



Summary

Annotation Validation Summary

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Total cells: 47,736

Total genes (HVG): 2000

Clustering resolution: 31 clusters

Finalized Cell Type Distribution:

CAF:	16,926	(35.5%)
CD8+ T:	6,910	(14.5%)
Myeloid:	5,005	(10.5%)
NK:	4,686	(9.8%)
CD4+ T:	3,663	(7.7%)
Malignant:	3,539	(7.4%)
Endothelial:	2,367	(5.0%)
Plasma:	1,984	(4.2%)
T reg:	1,524	(3.2%)
Unknown:	488	(1.0%)
T_CD8:	359	(0.8%)
T_reg:	169	(0.4%)
T_CD4:	89	(0.2%)
Myeloid_mono:	21	(0.0%)
Macrophage:	6	(0.0%)

Refinement Actions Performed:

- Resolved T/NK population (16,647 cells) into:
 - CD8+ T: 6,910 cells
 - CD4+ T: 3,663 cells
 - T reg: 1,524 cells
 - NK: 4,686 cells
- Validated marker expression patterns
- Confirmed cluster-specific marker specificity
- Generated dotplot and violin diagnostic plots

Output Files:

- dotplot_markers_vs_clusters.png
- violin_*.png (9 lineage-specific plots)
- dotplot_TNK_subclusters.png
- violin_TNK_*.png (4 T/NK subtype plots)
- annotation_validation_summary.txt
- annotation_refinement_summary.txt
- ovarian_cancer_annotated_refined.h5ad